

Bellabeat Smart Device Analysis Report

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Project Overview

This report presents a behavioral analysis of non-Bellabeat smart device usage to extract insights relevant for Bellabeat Leaf marketing and product positioning.

Data Loading

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.1.6
## v forcats    1.0.1      v stringr   1.6.0
## v ggplot2    4.0.1      v tibble    3.3.0
## v lubridate  1.9.4      v tidyr     1.3.2
## v purrr      1.2.0
```

```
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(stats)
```

```
data <- read_csv("final_dataset_prepared.csv")
```

```
## Rows: 964 Columns: 25
## -- Column specification -----
## Delimiter: ","
## chr   (4): activity_level, weekday, is_weekend, sleep_level
## dbl   (20): Id, TotalSteps, TotalDistance, TrackerDistance, LoggedActivitiesD...
## date  (1): ActivityDate
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

Descriptive Statistics

Global Metrics

```
summary_stats <- data %>%
  summarise(
```

```

    mean_steps = mean(TotalSteps, na.rm = TRUE),
    median_steps = median(TotalSteps, na.rm = TRUE),
    sd_steps = sd(TotalSteps, na.rm = TRUE),
    mean_active = mean(total_active_minutes, na.rm = TRUE),
    mean_sedentary_ratio = mean(sedentary_ratio, na.rm = TRUE),
    mean_sleep = mean(TotalMinutesAsleep, na.rm = TRUE),
    mean_time_in_bed = mean(TotalTimeInBed, na.rm = TRUE),
    mean_calories = mean(Calories, na.rm = TRUE)
  )
summary_stats

## # A tibble: 1 x 8
##   mean_steps median_steps sd_steps mean_active mean_sedentary_ratio mean_sleep
##   <dbl>         <dbl>    <dbl>      <dbl>          <dbl>      <dbl>
## 1     7492.         7280.    5120.        223.          0.802      420.
## # i 2 more variables: mean_time_in_bed <dbl>, mean_calories <dbl>

```

Weekday vs Weekend Comparison

```

weekday_analysis <- data %>%
  group_by(is_weekend) %>%
  summarise(
    mean_steps = mean(TotalSteps, na.rm = TRUE),
    mean_sleep = mean(TotalMinutesAsleep, na.rm = TRUE),
    mean_sedentary = mean(sedentary_ratio, na.rm = TRUE)
  )
weekday_analysis

## # A tibble: 2 x 4
##   is_weekend mean_steps mean_sleep mean_sedentary
##   <chr>         <dbl>    <dbl>          <dbl>
## 1 Weekday     7472.      414.          0.804
## 2 Weekend    7551.      436.          0.797

```

Behavioral Patterns

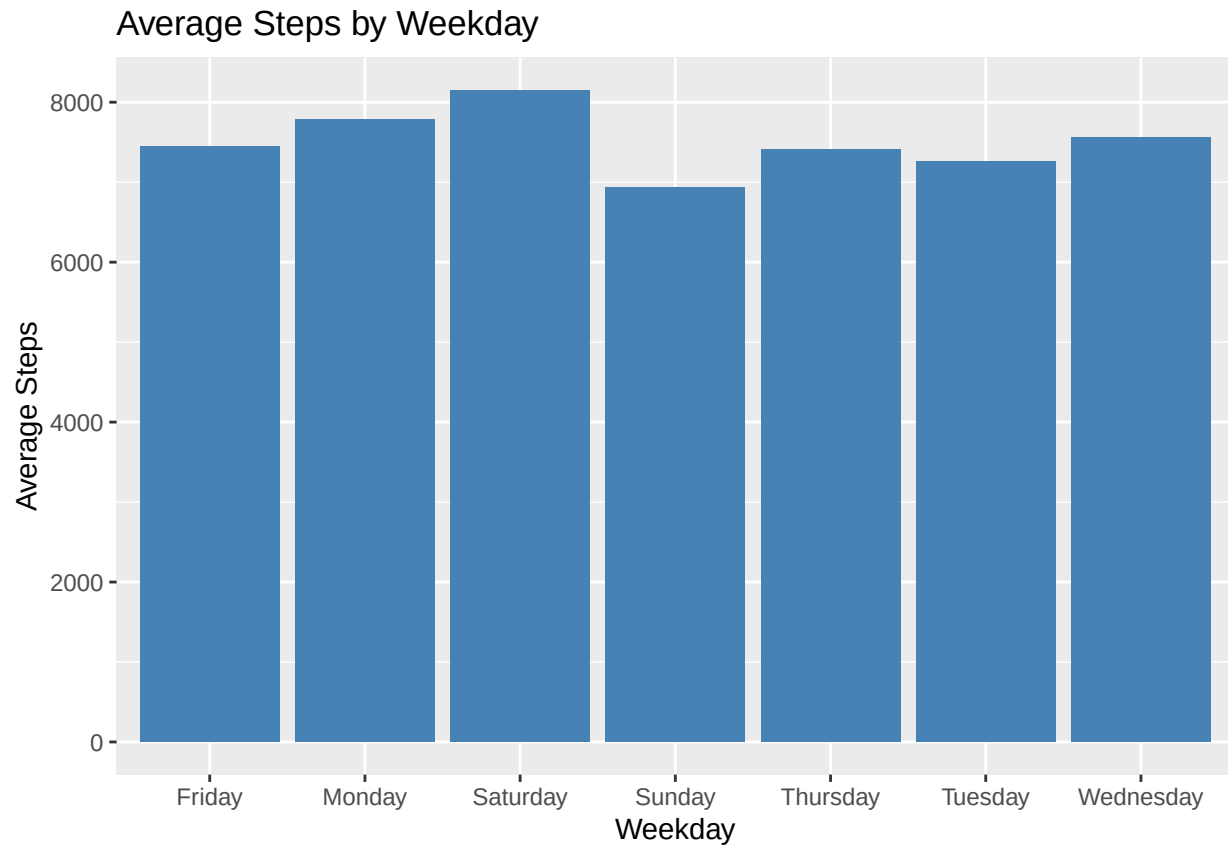
Average Steps by Weekday

```

avg_steps_weekday <- data %>%
  group_by(weekday) %>%
  summarise(mean_steps = mean(TotalSteps, na.rm = TRUE)) %>%
  arrange(match(weekday, c("Monday", "Tuesday", "Wednesday", "Thursday", "Friday", "Saturday", "Sunday")))

p_steps_weekday <- ggplot(avg_steps_weekday, aes(x = weekday, y = mean_steps)) +
  geom_bar(stat = "identity", fill = "steelblue") +
  ggtitle("Average Steps by Weekday") +
  xlab("Weekday") +
  ylab("Average Steps")
p_steps_weekday

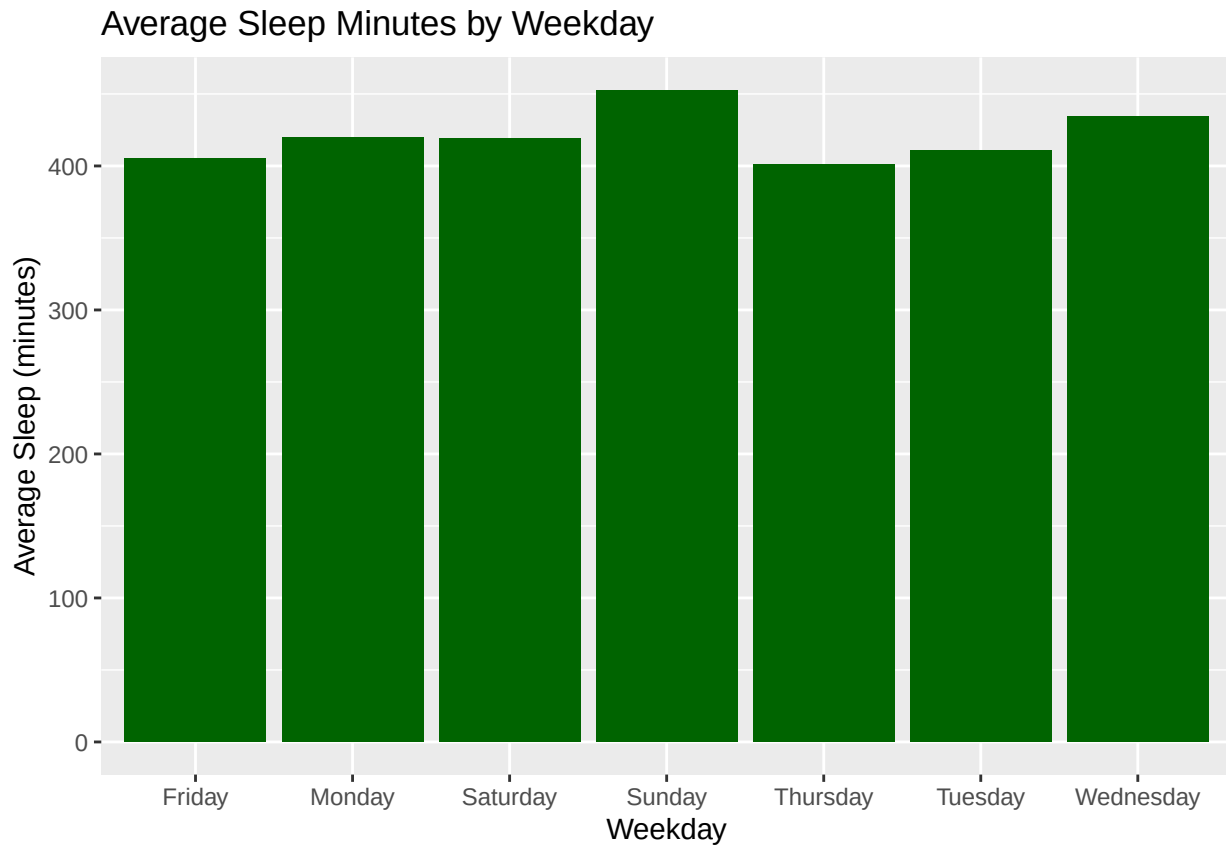
```



Average Sleep by Weekday

```
avg_sleep_weekday <- data %>%
  group_by(weekday) %>%
  summarise(mean_sleep_minutes = mean(TotalMinutesAsleep, na.rm = TRUE)) %>%
  arrange(match(weekday, c("Monday", "Tuesday", "Wednesday", "Thursday", "Friday", "Saturday", "Sunday"))))

p_sleep_weekday <- ggplot(avg_sleep_weekday, aes(x = weekday, y = mean_sleep_minutes)) +
  geom_bar(stat = "identity", fill = "darkgreen") +
  ggtitle("Average Sleep Minutes by Weekday") +
  xlab("Weekday") +
  ylab("Average Sleep (minutes)")
p_sleep_weekday
```



Correlation Analysis

Sleep vs Steps per User

```
user_corr <- data %>%
  group_by(Id) %>%
  summarise(
    cor_steps_sleep = if(sum(!is.na(TotalSteps) & !is.na(TotalMinutesAsleep)) > 1) {
      cor(TotalSteps, TotalMinutesAsleep, use = "complete.obs")
    } else {
      NA_real_
    }
  )
user_corr
```

```
## # A tibble: 33 x 2
##       Id cor_steps_sleep
##   <dbl>         <dbl>
## 1 1503960366    -0.325
## 2 1624580081     NA
## 3 1644430081    -0.401
## 4 1844505072     0.868
## 5 1927972279    -0.622
## 6 2022484408     NA
## 7 2026352035     0.184
```

```
## 8 2320127002      NA
## 9 2347167796    -0.205
## 10 2873212765     NA
## # i 23 more rows
```

Behavioral Segmentation

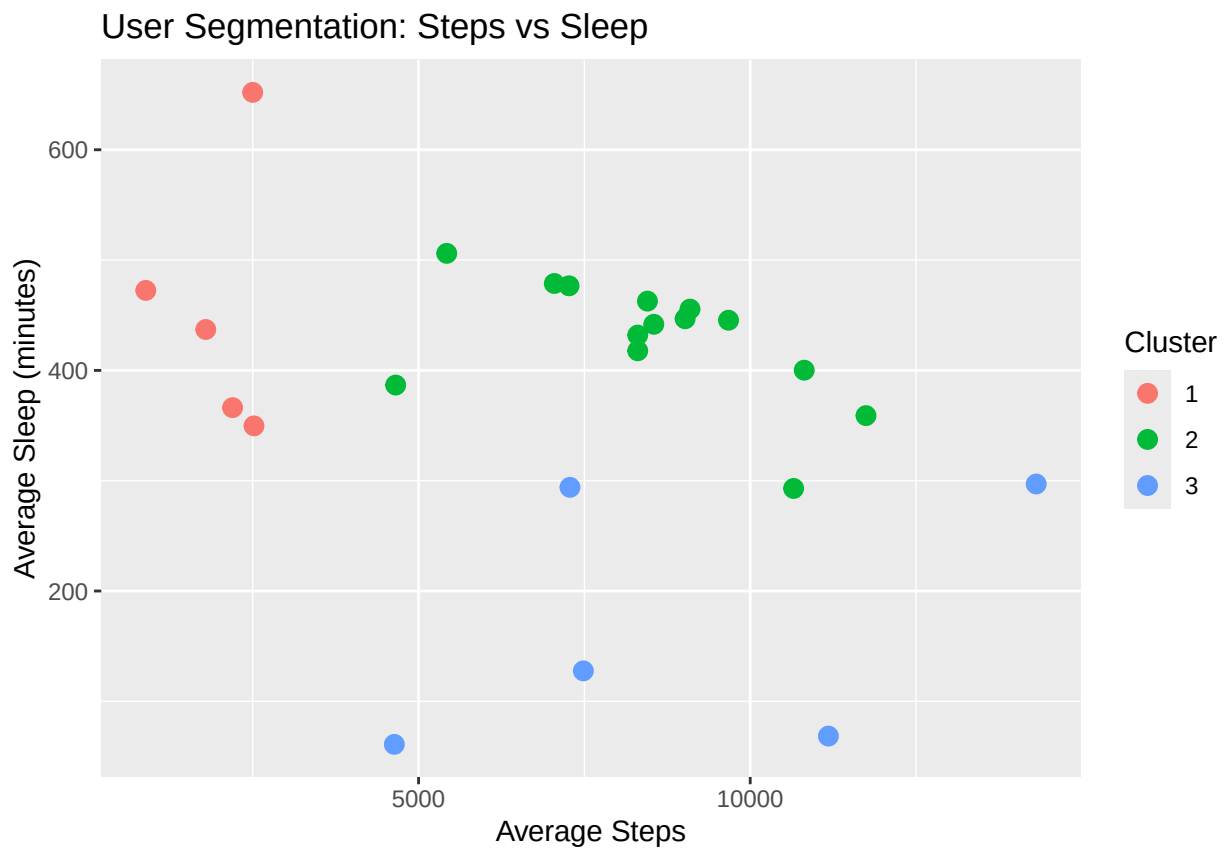
K-means Clustering

```
segmentation_vars <- data %>%
  group_by(Id) %>%
  summarise(
    avg_steps = mean(TotalSteps, na.rm = TRUE),
    avg_sleep = mean(TotalMinutesAsleep, na.rm = TRUE),
    avg_sedentary = mean(sedentary_ratio, na.rm = TRUE)
  ) %>%
  filter(!is.na(avg_steps) & !is.na(avg_sleep) & !is.na(avg_sedentary))

seg_scaled <- scale(segmentation_vars[, -1])
set.seed(123)
k_clusters <- kmeans(seg_scaled, centers = 3)
segmentation_vars$cluster <- k_clusters$cluster
segmentation_vars
```

```
## # A tibble: 24 x 5
##       Id avg_steps avg_sleep avg_sedentary cluster
##   <dbl>   <dbl>   <dbl>     <dbl>   <int>
## 1 1503960366 11745.    359      0.747     2
## 2 1644430081  7283.    294      0.847     3
## 3 1844505072  2499.    652      0.903     1
## 4 1927972279   888.    472.     0.969     1
## 5 2026352035  5425.    506.     0.723     2
## 6 2320127002  4635.     61      0.859     3
## 7 2347167796  9019.    447.     0.700     2
## 8 3977333714 10655.    293.     0.732     2
## 9 4020332650  2197.    366.     0.925     1
## 10 4319703577  7269.    477.     0.714     2
## # i 14 more rows
```

Clusters Visualization



Conclusions

- Users show clear weekday/weekend variations in steps and sleep.
- Clustering reveals distinct behavioral segments based on activity and sleep patterns.
- These insights can inform marketing strategies and product positioning for Bellabeat Leaf.